



Powers and roots using a calculator

Task A: Evaluate a number using a square root

1) **Evaluate** the following. If your answer is a decimal, write all the digits from your calculator display.

a) **Evaluate** $3.45^2 = 11.9025$

b) **Evaluate** $1.2^2 + 13.4^2 = 181$

c) **Evaluate** $\sqrt{76.7376} - 2.3^2 = 3.47$

d) **Evaluate** $\sqrt{9^2 + 12^2} = 15$

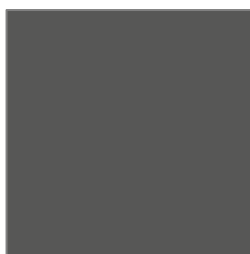
2) Using a calculator, work out the missing lengths of the following shaded areas. Each shape is a square.



Area = 127.69 cm^2

Length = $\sqrt{127.69}$

Length = 11.3 cm



Area = 4.6225 m^2

Length = $\sqrt{4.6225}$

Length = 2.15 m



Area = 0.000001 m^2

Length = $\sqrt{0.000001}$

Length = 0.001 m

3) Using a calculator, explain why the following evaluate to the answer given.

a) $\sqrt{\sqrt{\sqrt{65536}}} = 4$

This means the square root of the square root of the square root of 65 536.

$\sqrt{65536} = 256$

$\sqrt{256} = 16$

$\sqrt{16} = 4$

b) $\sqrt{53^2} = 53$

Square and square root are inverses, thus they cancel each other out.



Task B: Evaluate using exponents

1) **Evaluate** the following. If your answer is a decimal, write all the digits from your calculator display.

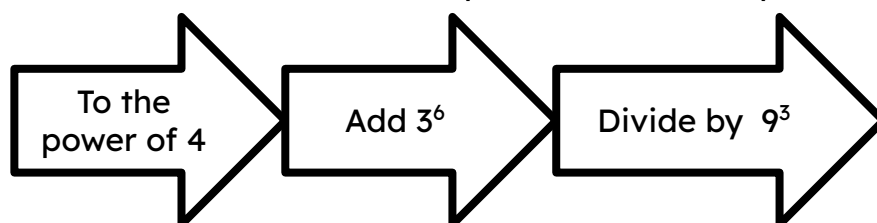
a) **Evaluate** $4^6 - 6^4 = 2800$

b) **Evaluate** $27^4 \div 9^2 \div 3^7 = 3$

c) **Evaluate** $4.2^3 \div 2^4 = 4.6305$

d) **Evaluate** $\sqrt{3^4 + 4^3 + 5^2 - 1^7} = 13$

2) Sofia followed this function machine. Her input was 27. Her output was 730.



Which of the following buttons did she press to get the correct answer?

a



b



Task C: $\times 10^x$ button on a calculator

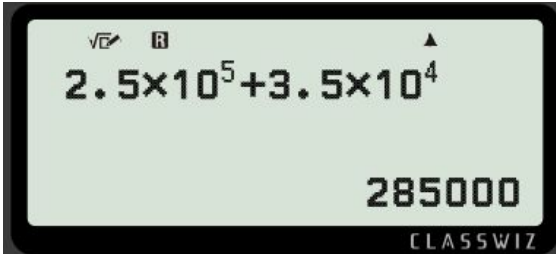
 1) **Evaluate** the following:

a) $1.7 \times 10^4 = 17\,000$

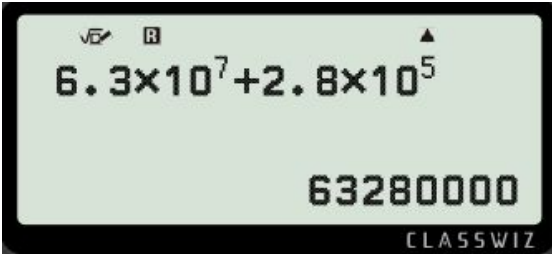
b) $3.9 \times 10^8 - 2.6 \times 10^7 = 364\,000\,000$

c) $1.2 \times 10^2 + 1.2 \times 10^3 + 1.2 \times 10^4 = 13\,320$

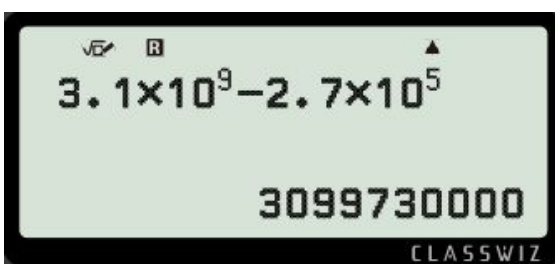
2) Work out the missing exponents.

a) 

A calculator screen showing the expression $2.5 \times 10^5 + 3.5 \times 10^4$ and the result 285000. The screen also displays icons for square root, fraction, and a triangle.

b) 

A calculator screen showing the expression $6.3 \times 10^7 + 2.8 \times 10^5$ and the result 63280000. The screen also displays icons for square root, fraction, and a triangle.

c) 

A calculator screen showing the expression $3.1 \times 10^9 - 2.7 \times 10^5$ and the result 3099730000. The screen also displays icons for square root, fraction, and a triangle.